ENGAGE TRACK

Multi-Agents for Presence Recognition& Intelligent Monitoring LMS

B. Tech Major Project Synopsis — 2026–2027

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Abstract

Engage Track is an AI-powered online classroom monitoring system that uses computer vision and multi-agent technology to improve student engagement and attendance management. The system verifies student presence, monitors attentiveness through facial and head-pose analysis, and detects possible identity changes during live sessions. Built using OpenCV, Media Pipe, and Python, Engage Track provides real-time insights to educators while promoting academic integrity, active participation, and responsible use of AI in online learning environments

Problem Statement

Current online learning platforms can record attendance but cannot accurately determine whether a student is genuinely present, attentive, or being represented by someone else during a live session. This lack of real-time monitoring reduces student engagement and creates challenges in maintaining academic integrity. Therefore, there is a need for an intelligent system that can automatically verify student presence, analyze engagement levels, and detect identity changes during online classes using computer vision and AI techniques. This leads to issues such as low engagement, proxy attendance, and reduced academic integrity. Engage Track addresses these challenges using AI and computer vision to monitor student presence and engagement in real time.

Objectives

The primary objective of Engage Track is to develop an AI-powered system that can automatically verify student attendance, monitor engagement levels, and detect identity changes during online classes. The project aims to enhance academic integrity, improve student participation, and provide educators with real-time insights into classroom engagement using computer vision and multi-agent AI technologies. The system aims to automatically verify student attendance by ensuring that the enrolled participant remains present throughout the session. It also focuses on monitoring student engagement by analyzing facial expressions, head movements, and attention levels in real time. Another key objective is to detect identity changes or proxy attendance during live classes to support academic integrity. Additionally, the project seeks to provide teachers with meaningful insights into student participation and classroom engagement while maintaining a fair, ethical

System Architecture

1. User Authentication Module

This module authenticates students and faculty members using login credentials and OTP/email verification. It ensures that only authorized users can access the online classroom and system resources.

2. Video Conferencing Module

This module provides the live virtual classroom environment using an existing video conferencing platform (WebRTC/SDK). It captures real-time video streams for further analysis.

3. Computer Vision Module

This module processes live video feeds using OpenCV and MediaPipe. It detects faces, extracts facial landmarks, and tracks head movements to gather data about student presence and attention.

4. Attendance Agent Module

The Attendance Agent continuously monitors whether the registered student remains visible during the session. It automatically records attendance and identifies prolonged absence from the camera.

5. Engagement Agent Module

This module analyzes facial orientation, eye focus, and head pose to determine student attentiveness. It identifies disengagement and generates alerts when attention levels drop.

6. Integrity Agent Module

The Integrity Agent verifies student identity throughout the session. It detects possible identity changes or proxy attendance attempts and flags suspicious activities for review.

7. Database Management Module

This module stores user details, attendance records, engagement reports, session logs, and alert information securely using SQLite or Firebase.

8. Dashboard & Reporting Module

This module provides teachers and administrators with attendance statistics, engagement analytics, integrity alerts, and performance reports through an interactive dashboard.

Technology Stack

Component	**Technology**
AI / LLM Runtime	Ollama (Qwen 2.5, Llama 3.2, Mistral)
Backend & Database	Supabase Postgres
Styling and UI	Tailwind CSS, Lucide React
External APIs	FastAPI, SMTP
Frontend Framework	React with TypeScript

Expected Outcomes

The expected outcome of Engage Track is a functional AI-powered system capable of automatically verifying student attendance, monitoring engagement levels, and detecting identity changes during online classes. The system will provide real-time insights into student participation, helping educators improve classroom interaction and maintain academic integrity. It is expected to reduce proxy attendance, enhance learning effectiveness, and demonstrate the practical application of computer vision and multi-agent AI technologies in online education.

Conclusion

Engage Track is an innovative AI-powered online classroom monitoring system that combines Computer Vision and Multi-Agent Intelligence to enhance the quality of virtual learning. The system effectively verifies student attendance, monitors engagement levels, and detects identity changes in real time. By providing accurate participation insights and supporting academic integrity, Engage Track helps educators create a more interactive, secure, and accountable online learning environment. The project demonstrates the practical application of modern AI technologies in education and contributes to improving the effectiveness of digital classrooms Furthermore, Engage Track promotes academic integrity by detecting suspicious activities such as identity changes and unauthorized participation. The use of technologies such as OpenCV, Media Pipe, and Python demonstrates the practical application of AI in solving real-world educational problems. Overall, the project contributes to creating a more interactive, secure, and accountable online learning environment while showcasing the potential of intelligent monitoring systems in modern digital education.